

FINAL PROJECT REPORT

Medical Diagnosis Center by ServiceNow Service Portal

Project Name: Medical Diagnosis Center by ServiceNow Service Portal

Platform: ServiceNow

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1. INTRODUCTION

1.1 Project Overview

The Medical Diagnosis Center by ServiceNow Service Portal is a ServiceNow-based healthcare-service workflow prototype focused on diagnostic-test discovery, appointment booking, appointment tracking and diagnostic report access. The patient-facing experience is delivered through Service Portal pages/widgets and connects the Medical Test Catalog with a structured Appointment Booking journey.

The demonstrated solution uses ServiceNow-native configuration and data structures rather than an external frontend/backend stack. Patient and diagnostic-test information is maintained in structured tables, appointment transactions are stored in the appointment table, report information is maintained in the report table, and business rules automate key lifecycle actions.

1.2 Purpose

- Provide a centralized ServiceNow portal for diagnostic-test discovery and appointment booking.
- Reduce manual coordination by standardizing patient information and appointment capture.
- Give patients a clear booking journey: Details → Schedule → Payment → Done.
- Provide a single place to track appointments and access available lab reports.
- Use ServiceNow tables and business rules to maintain a traceable appointment/report lifecycle.
- Provide a maintainable ServiceNow implementation that can be extended with integrations and analytics.

2. IDEATION PHASE

2.1 Problem Statement

Patients often need a simple way to discover diagnostic tests, understand basic test information, book a suitable appointment, track the booking, and access completed reports. When these activities are handled through fragmented or manual communication, patients may repeat information, have limited visibility of appointment status, and depend on staff for follow-up. The operational team also needs structured records for patients, tests, appointments, and reports.

2.2 Empathy Map Canvas

Stakeholder	Think & Feel	See	Say & Do	Pain	Gain
Patient	Wants a simple and predictable booking experience.	Portal, test catalog, booking stages, appointment/report pages.	Searches tests, books, checks appointments, opens reports.	Repeated coordination, unclear status, difficulty accessing reports.	Guided booking, visibility, centralized access.
Diagnostic Staff	Wants complete appointment information and clear operational status.	Appointment records and patient/test details.	Reviews and fulfils scheduled diagnostic services.	Incomplete information and manual follow-up.	Structured records and cleaner handoff.
ServiceNow Administrator	Wants maintainable configuration and reliable automation.	Tables, portal pages/widgets, business rules, ACLs.	Configures and monitors application behaviour.	Troubleshooting and inconsistent configuration.	Centralized configuration and traceability.
Organization	Wants an efficient digital diagnostic service workflow.	Portal, appointment records and reports.	Provides diagnostic services through the configured workflow.	Fragmented service handling and limited visibility.	Standardized digital service lifecycle.

2.3 Brainstorming

- Searchable Medical Test Catalog with test description, category, duration, availability and price.
- Book Now action that starts a structured appointment journey.
- Multi-stage appointment booking: Details → Schedule → Payment → Done.
- Patient and diagnostic-test master data maintained in ServiceNow tables.
- Appointment records with status, scheduled date, slot time, patient and rejection information.
- Business-rule automation for checkout processing, appointment completion/report creation, and rejection-related updates.
- My Appointments page for tracking booking history and status.
- My Lab Reports page for accessing diagnostic reports.
- Role/ACL-based protection and ServiceNow administrative configuration.
- Future integration with laboratory/hospital systems, messaging, analytics and carefully governed AI assistance.

The selected concept is a centralized ServiceNow Medical Diagnostic Center because it directly connects diagnostic-test discovery, booking, appointment tracking and report access in one platform workflow.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Patient Experience	System Response
Need Test	Patient identifies a diagnostic need.	Portal provides a centralized Medical Test Catalog.
Browse/Search	Patient searches or browses available tests.	Catalog presents diagnostic test information and Book Now.
Select Test	Patient chooses the required test.	Selected test becomes the basis for the booking flow.
Details	Patient enters required information.	Booking captures patient details.
Schedule	Patient selects appointment schedule/slot.	Appointment information is prepared for storage.
Payment	Patient proceeds through payment stage.	Booking advances to the final stage.
Done	Patient completes the booking.	Appointment record is stored/updated.
Track	Patient opens My Appointments.	Appointment status and details are displayed.
Complete	Diagnostic service is completed.	Completion-related business rule can trigger report processing.
Report	Patient opens My Lab Reports.	Available report information is presented.

3.2 Solution Requirements

Requirement	Description
FR-01	Provide a ServiceNow Service Portal Medical Diagnostic Test Center home page.
FR-02	Provide a Medical Test Catalog with search/browse and Book Now capability.
FR-03	Display configured diagnostic-test information such as description, category, duration, availability and price.
FR-04	Provide an Appointment Booking flow with Details, Schedule, Payment and Done stages.

FR-05	Capture and maintain patient information required for appointment processing.
FR-06	Create and maintain appointment records with patient, schedule, status and related fields.
FR-07	Provide My Appointments for appointment history, status and details.
FR-08	Create/update diagnostic report information when the appointment reaches the configured completion condition.
FR-09	Provide My Lab Reports for patient-facing report access.
FR-10	Apply business-rule automation for checkout, completion/report creation and rejection-related updates.
FR-11	Use ServiceNow roles and ACLs to control access to configured records.
FR-12	Support administrative troubleshooting through ServiceNow configuration and execution/record information.

Non-Functional Requirements

Category	Requirement
Usability	The patient journey should be simple, clear and understandable.
Security	Patient, appointment and report data must be protected using the configured ServiceNow access model.
Reliability	Booking and business-rule processing should execute consistently.
Performance	Portal pages and booking interactions should respond without abnormal delay in normal project testing.
Maintainability	Tables, widgets/pages and business rules should remain understandable and configurable.
Data Integrity	Patient, test, appointment and report relationships should remain consistent.
Auditability	Appointment status and configured record fields should provide traceability.

3.3 Data Flow Diagram

Level 0 context: Patient → Medical Diagnostic Test Center Service Portal → Diagnostic Test / Appointment / Report Services → Diagnostic Staff / ServiceNow Administrator.

- Patient enters portal → searches Medical Test Catalog → selects test → Book Now
- → Appointment Booking: Details → Schedule → Payment → Done
- → Appointment Table stores appointment information
- → Business Rules process checkout/completion/rejection-related logic
- → Diagnostic service completion
- → Report Table stores report information
- → My Lab Reports presents available report information

3.4 Technology Stack

Component	Technology	Purpose
User Interface	ServiceNow Service Portal / Portal Pages / Widgets	Patient-facing diagnostic service experience.
Diagnostic Catalog	Diagnostic Test Table + Catalog UI	Test discovery, details, availability and Book Now.
Booking	Appointment Booking page/widget	Details → Schedule → Payment → Done.
Patient Data	Patient Table	Patient identity and contact/demographic information.
Appointment Data	APPOINTMENT TABLE	Appointment status, slot, date, patient and related operational fields.
Report Data	REPORT TABLE	Report date/number, verified-by, appointment and patient relationships.
Automation	ServiceNow Business Rules	Checkout processing, report creation on completion and rejection-related updates.
Access Control	Roles + ACLs	Controlled access to configured application data.
Platform	ServiceNow instance / PDI	Execution, configuration and data storage environment.

4. PROJECT DESIGN

4.1 Problem Solution Fit

- Customer segment: patients, diagnostic operational staff, ServiceNow administrators and the diagnostic-center organization.
- Jobs-to-be-done: discover tests, book appointments, track appointments, complete diagnostic services and access reports.
- Root causes: fragmented communication, repeated information capture, limited appointment visibility and disconnected report access.
- Constraints: the current implementation is a ServiceNow project prototype; direct hospital/EMR/laboratory integration is not demonstrated.
- Solution fit: Service Portal + Medical Test Catalog + Appointment Booking + structured tables + business rules + My Appointments + My Lab Reports + security.

4.2 Proposed Solution

The proposed solution is an end-to-end ServiceNow Medical Diagnostic Center. A patient enters the portal, discovers a diagnostic test, starts booking, supplies patient information, selects a schedule, proceeds through payment, and completes the booking. The resulting appointment is maintained in ServiceNow, while configured business rules support lifecycle automation. Patients can then track appointments and access available diagnostic reports.

Novelty: the project applies ServiceNow portal, structured application data and low-code automation to create a connected diagnostic-service journey rather than treating the platform only as a record repository.

Customer impact: simpler test discovery, a guided booking sequence, improved appointment visibility and easier report access.

Operational impact: structured records, automation, controlled access and traceable lifecycle processing.

4.3 Solution Architecture

Architecture layers: User Layer (Patient) → ServiceNow Experience Layer (Service Portal, Medical Test Catalog, Appointment Booking) → Data Layer (Patient, Diagnostic Test, Appointment and Report tables) → Automation Layer (Business Rules) → Tracking & Reporting Layer (My Appointments and My Lab Reports) → Administration/Security Layer (Roles, ACLs and ServiceNow configuration).

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Epic / Focus	Outcome
Sprint 1	Portal & Medical Test Catalog	Portal home, test catalog, test data, search/browse and Book Now.
Sprint 2	Appointment Booking & Data	Details, Schedule, Payment, Done flow; patient and appointment data.
Sprint 3	Automation & Patient Tracking	Business rules, My Appointments, completion/report processing and My Lab Reports.
Sprint 4	Testing, Refinement & Documentation	Functional/performance checks, access validation, screenshots and project documents.

The planning model uses four relative implementation sprints: portal/catalog foundation, booking/data, automation/tracking, and testing/documentation. Exact historical calendar dates are not claimed by this report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional and Performance Testing

Test ID	Test Scenario	Final Result
FT-01	Portal home and navigation	Pass
FT-02	Medical Test Catalog search/browse	Pass
FT-03	Diagnostic test information display	Pass
FT-04	Book Now starts appointment booking	Pass
FT-05	Patient Details stage	Pass
FT-06	Schedule stage	Pass
FT-07	Payment stage	Pass
FT-08	Done / booking completion	Pass
FT-09	Appointment record creation/update	Pass
FT-10	My Appointments tracking	Pass
FT-11	Appointment completion/report automation	Pass
FT-12	My Lab Reports access	Pass
FT-13	Roles / ACL access behaviour	Pass
FT-14	End-to-end patient workflow	Pass
PT-01	Portal/form response behaviour	Pass – no abnormal delay reported in project testing
PT-02	Repeated booking/workflow consistency	Pass
PT-03	Business-rule execution consistency	Pass

AI/ML, GenAI, Power BI and Tableau model-performance metrics are not applicable to the current demonstrated implementation because the project does not use a trained AI/ML model or BI platform. No unsupported metrics are invented.

7. RESULTS

7.1 Output

The final implementation demonstrates a connected ServiceNow diagnostic-service experience from test discovery through appointment booking, tracking and report access.

Output Area	Implemented Result
Portal	Medical Diagnostic Test Center portal with navigation to catalog, appointments and reports.
Catalog	Diagnostic test catalog with search/browse, information and Book Now actions.
Booking	Four-stage Details → Schedule → Payment → Done journey.
Appointment	Structured appointment records with status, schedule, patient and related fields.
Automation	Business rules for checkout, completion/report creation and rejection-related updates.
Tracking	My Appointments provides appointment visibility.
Reporting	My Lab Reports provides access to diagnostic report information.
Security	Configured ServiceNow access controls protect application records.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Current Limitations
Structured booking: Standardizes the patient journey instead of relying on repeated manual coordination.	Prototype scope: Current implementation is a project prototype, not a production clinical information system.
Centralized information: Patient, test, appointment and report information are maintained within the ServiceNow solution.	External integration: Direct hospital/EMR/EHR/laboratory/API integration is not demonstrated.
Clear lifecycle: Details → Schedule → Payment → Done creates a predictable booking path.	Clinical decision support: The system does not autonomously diagnose or prescribe.
Automation: Business rules reduce repetitive appointment/report processing.	Advanced analytics: Production-grade SLA, escalation, dashboards and BI are future scope.
Visibility: Patients can review appointments and available reports.	Messaging: External SMS/WhatsApp and similar channels are future scope.
Maintainability: ServiceNow configuration and low-code mechanisms support incremental changes.	AI/ML metrics: No trained AI/ML model is part of the current demonstrated implementation, so model-performance metrics are not applicable.

9. CONCLUSION

The Medical Diagnosis Center by ServiceNow Service Portal project demonstrates how ServiceNow can provide a structured digital workflow for diagnostic-test discovery, appointment booking, appointment tracking and report access. The implementation connects the patient-facing portal with structured ServiceNow data and automation so that the booking lifecycle is visible and maintainable.

The main implementation focus is reliable service handoff: presenting useful test information, capturing patient details, guiding the booking through Details → Schedule → Payment → Done, storing appointment information, applying configured business rules, and providing My Appointments and My Lab Reports experiences. The prototype does not claim autonomous clinical diagnosis or treatment.

10. FUTURE SCOPE

- Hospital, EMR/EHR and laboratory integration through authorized APIs.
- Automated task assignment, fulfilment tracking and closure for diagnostic operations.
- Advanced SLA tracking, escalation, dashboards and operational analytics.
- Email/SMS/WhatsApp or other approved notification channels.
- AI-assisted test/service classification, anomaly detection, prioritization or intelligent routing with healthcare governance.
- Patient history and longitudinal service analytics where legally and operationally appropriate.
- Production deployment hardening, monitoring, audit and performance benchmarking.

11. APPENDIX

Appendix A – Functional Data Mapping

Data Element / Table	Purpose	Use in Workflow
Patient	Patient information	Used during booking and appointment relationship.
Diagnostic Test	Test definition	Description, category, duration, availability and price-related information.
Appointment	Booking transaction	Patient, status, scheduled date, slot time and rejection-related information.
Report	Diagnostic result record	Report date/number, verification, appointment and patient relationship.
Business Rules	Automation	Checkout, completion/report creation and rejection-related updates.

Appendix B – Current Implementation Boundary

Implemented / Current Scope: Service Portal, Medical Test Catalog, Appointment Booking, Patient and Diagnostic Test data, Appointment and Report tables, My Appointments, My Lab Reports, business-rule automation, roles/ACLs and administrative configuration.

Future / Not Demonstrated: direct hospital/EMR/EHR/laboratory APIs, external messaging channels, advanced BI/analytics, trained AI/ML clinical models, autonomous diagnosis or treatment recommendations, and production-scale benchmarking.